Causal Inference: Retention Offer Use Case

Chapter 13: Standardization and the Parametric G-Formula

Outcome Regression Modeling, Empirical Standardization, and Comparison with IP Weighting

Applied Causal Inference Series

October 12, 2026

Executive Summary

Chapters 11 and 12 introduced model-based causal inference, demonstrating how **Inverse Probability (IP) Weighting** models the treatment mechanism $f(A \mid L)$ (the propensity score) to construct a pseudo-population where confounders L are unassociated with treatment A.

Chapter 13 presents the primary alternative approach to conditional adjustment: **Standardization** (also known as the **Parametric G-Formula**). Rather than modeling who receives treatment, standardization models the **outcome mechanism** $E[Y \mid A, L]$. Under conditional exchangeability, positivity, and consistency, the parametric g-formula fits an outcome regression model $\hat{E}[Y \mid A, L]$ and marginalizes (averages) predicted counterfactual outcomes across the empirical distribution of baseline customer covariates L.

1 Nonparametric vs. Parametric Standardization

1.1 Nonparametric Standardization

In a discrete setting with a small vector of confounders L, the standardized risk under treatment level a is computed by taking a weighted average of conditional outcome risks across confounder strata l:

$$\Pr[Y^a = 1] = \sum_l \Pr[Y = 1 \mid A = a, L = l] \times \Pr[L = l] \tag{1}$$

1.2 The High-Dimensional Breakdown

In enterprise Customer Relationship Management (CRM) databases, baseline covariates $L = (L_1, L_2, \ldots, L_p)$ include continuous variables (customer tenure, monthly bill, data usage) and multi-category traits (support ticket history, plan type).

- When L is high-dimensional or continuous, most confounder strata l contain zero or very few subscribers.
- Nonparametric conditional means $\hat{\Pr}[Y = 1 \mid A = a, L = l]$ cannot be computed (0/0).
- **Solution**: Replace nonparametric stratum-specific means with a **parametric outcome regression model** $\hat{E}[Y \mid A, L]$.

2 Algorithm for the Parametric G-Formula

The parametric g-formula for a binary outcome (e.g., customer churn $Y \in \{0, 1\}$) and treatment (retention offer $A \in \{0, 1\}$) proceeds in three distinct computational steps:

2.1 Step 1: Fit the Outcome Regression Model

Using the observed dataset of n subscribers, fit a parametric model for the conditional expectation $E[Y \mid A, L]$ (e.g., a logistic regression model):

$$\text{logit } \Pr[Y = 1 \mid A, L] = \theta_0 + \theta_1 A + \boldsymbol{\theta}_2^T L + \theta_3 (A \times L) \tag{2}$$

where θ_1 captures the main effect of the offer, $\boldsymbol{\theta}_2$ captures baseline covariate risk, and θ_3 allows for treatment effect modification.

2.2 Step 2: Predict Counterfactual Outcomes

For **every single subscriber** i in the database ($i = 1, \ldots, n$), construct two counterfactual datasets:

1. **Force** $A_i = 1$: Predict $\hat{Y}_i^{a=1} = \hat{\Pr}[Y = 1 \mid A = 1, L_i]$ using subscriber i's actual baseline covariates L_i.
2. **Force** $A_i = 0$: Predict $\hat{Y}_i^{a=0} = \hat{\Pr}[Y = 1 \mid A = 0, L_i]$ using subscriber i's actual baseline covariates L_i.

2.3 Step 3: Empirical Standardization (Averaging)

Average these predicted counterfactual outcomes across the entire sample of n subscribers:

$$\widehat{\Pr}[Y^{a=1} = 1] = \frac{1}{n} \sum_{i=1}^{n} \hat{\Pr}[Y = 1 \mid A = 1, L_i] \tag{3}$$

$$\widehat{\Pr}[Y^{a=0} = 1] = \frac{1}{n} \sum_{i=1}^{n} \hat{\Pr}[Y = 1 \mid A = 0, L_i] \tag{4}$$

The estimated **Average Causal Risk Difference (RD)** is:

$$\widehat{RD} = \widehat{\Pr}[Y^{a=1} = 1] - \widehat{\Pr}[Y^{a=0} = 1] \tag{5}$$

3 CRM Applied Use Case: Evaluating Retention Offers

3.1 Dataset Context

A telecom firm evaluates a retention discount $A \in \{0, 1\}$ targeted at subscribers. The CRM database tracks continuous tenure L_1 (months), monthly spend L_2 (\$), and support tickets L_3. High-risk, low-tenure customers were disproportionately targeted.

3.2 Model Specification & Results

1. **Outcome Model**:

$$\text{logit } \Pr[Y = 1 \mid A, L] = -1.2 - 0.65A - 0.04L_1 + 0.01L_2 + 0.35L_3 - 0.02(A \times L_1)$$

2. **Counterfactual Predictions**:

- $\widehat{\Pr}[Y^{a=1} = 1] = 0.142$ (14.2% projected churn if ALL subscribers receive the offer).
- $\widehat{\Pr}[Y^{a=0} = 1] = 0.228$ (22.8% projected churn if NO subscribers receive the offer).

3. **Causal Effect Estimates**:
   - **Risk Difference (RD)**: $0.142 - 0.228 = -0.086$ (8.6 percentage point absolute reduction in churn).
   - **Risk Ratio (RR)**: $0.142/0.228 = 0.623$ (37.7% relative churn reduction).

4 IP Weighting vs. Standardization: A Comparative Analysis

Feature	Inverse Probability (IP) Weighting	Standardization (Parametric G-Formula)
Primary Model Target	Treatment Mechanism $f(A \mid L)$ (Propensity Score)	Outcome Mechanism $E[Y \mid A, L]$
Positivity Sensitivity	**High**: Extremely small propensity scores yield huge weights W^A, inflating variance.	**Low / Hidden**: Predicts via model extrapolation; avoids wild weights but risks structural misspecification.
Effect Modification	Requires fitting Moderated MSMs $SW^A(V)$.	Straightforward: Include interaction terms $(A \times V)$ in outcome regression.
Continuous Treatments	Requires density estimation $f(A \mid L)$ for weights.	Models response curves directly $E[Y \mid A = a, L]$ across continuous dosage a.
Model Requirement	Unbiased if $f(A \mid L)$ is correctly specified.	Unbiased if $E[Y \mid A, L]$ is correctly specified.

Table 1: Methodological comparison of Inverse Probability Weighting and Standardization.

5 Doubly Robust Estimation: The Best of Both Worlds

When modeling observational data, we face a dilemma:

- If our propensity score model $f(A \mid L)$ is misspecified, IP weighting is biased.
- If our outcome regression model $E[Y \mid A, L]$ is misspecified, standardization is biased.

5.1 Doubly Robust (DR) Estimator

A **Doubly Robust Estimator** (e.g., Augmented IPW or Targeted Maximum Likelihood Estimation / TMLE) combines both models:

$$\widehat{E}[Y^a]_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{I(A_i = a)Y_i}{\hat{f}(A_i \mid L_i)} - \frac{I(A_i = a) - \hat{f}(a \mid L_i)}{\hat{f}(a \mid L_i)}\hat{E}[Y \mid A = a, L_i]\right] \tag{6}$$

5.2 The Doubly Robust Property

The estimator $\widehat{E}[Y^a]_{DR}$ is **consistent for the true counterfactual mean** $E[Y^a]$ if **EITHER** the propensity score model $\hat{f}(A \mid L)$ **OR** the outcome regression model $\hat{E}[Y \mid A, L]$ is correctly specified.

Summary Checklist for Applied Analysts

1. **Choose the Nuisance Model**: Decide whether treatment assignment $f(A \mid L)$ or outcome determination $E[Y \mid A, L]$ is easier to model with domain knowledge.
2. **Check Positivity**: If propensity scores are near 0 or 1, standardization or doubly robust estimation is preferred over standard IP weighting.
3. **Use Bootstrap for CIs**: Obtain standard errors and 95% confidence intervals for g-formula estimates by nonparametrically bootstrapping the entire 3-step pipeline.
4. **Consider Doubly Robust Estimators**: In critical business or clinical decisions, deploy DR estimators to protect against single-model misspecification.